Companion in the Boundary Problem network. Build companions: How to Build a UFO · The DIY Hoverbike · The Hexalock. Mechanism: The Shape of Gravity · Warp-Drive Math Audit. Force law: χν → Force.

The Hover Disk

***The substrate-selection paper** — which material can hold the most coherently-stored energy per kg, in the right geometry. The four material levers, and where a real body sits on the coherence dial $S_{coh}{\cdot}f_{cons}$. Mechanism questions live in `the_shape_of_gravity.html` and `WARP_DRIVE_MATH_AUDIT.md`, not here.*

Osika & collaborators — draft 0.4, 2026-07-19 (0.3: 2026-06-19).

> ***Role of this paper: substrate selection.** This is the materials-ranking paper — which material can hold the most coherently-stored energy per kg, in the right geometry. The mechanism story lives elsewhere: the derived scalar elastic-metric law and what it closes (weak-field gravity, factor-2 light bending included, scalar-only) is `the_shape_of_gravity.html`; the force-law symbol table, the two-regime conservation audit, the polarizable-vacuum floor, and the coherence dial are `WARP_DRIVE_MATH_AUDIT.md`. Send mechanism questions there. A hover disk lifts by the substrate-coupling channel (per-vertex sharing strength $g \le 1/\sqrt{3}$, aggregated as $x_{sub} = \Sigma\, g_v\, \psi^*\psi$; earlier drafts called this x_v), not by pushing air — so its lift is a materials problem. The coupling splits into **four material levers**: the **energy ceiling** (specific resilience $\sigma^2/(E\rho)$), **coherence** (Q and the coherent fraction S_{coh}), **consonance** (geometric match to the lattice — fidelity f_{cons}, ceiling $e^{-P/12} \approx 0.873$), and the **coupling efficiency η**. The first three are sound material levers; only the frequency-scaling of η is genuinely open — the **A/B fork** of §4 (hypothesis A: $\eta \approx v_{phase}/c$, so EM drive wins, acoustic $\lesssim 10^{-5}$ vs optical ~0.1–0.5; hypothesis B: the coupling is set by the stored field, DC suffices), which the §7 bench settles. On the energy ceiling bronze loses while **single-crystal diamond is the strongest single first core** — structural champion and high-index optical substrate; no single material carries all four UFO §8 predicates, so the convergent body is a **layered stack (§8)**. **Honest status:** the force law is derived in form; its magnitude — $C_{sub}\cdot|x_{sub}|^2$ — is open and unmeasured. The bare-vacuum value is floored at the Einstein constant $8\pi G/c^4 \approx 2\times10^{-43}$ (`WARP_DRIVE_MATH_AUDIT.md` §8); the claim that coherence lifts a real body far above that floor is **the single open bet**. The nulls that bound the coupling are **published literature nulls** (Tajmar 2004; Bartlett & Tajmar 2021; Talley 1991, §15) — **the project's own bench has not yet measured any substrate**. This paper ranks the substrates that would maximise the four levers; it*

> *does not claim lift — it names what to build and measure, and where a real body lands on the coherence dial $P_w = (S_{coh}{\cdot}f_{cons}){\cdot}\kappa{\cdot}½\varepsilon_r\varepsilon_0E^2$ (a hover disk needs only $\sim10^{-6}$ of full coherence to self-lift). Reproducible numbers:* `hover/disk_material_ladder.py` .

> **Companion papers — and the division of labour.** This paper's enduring job is **substrate selection**; the **mechanism** story lives in `the_shape_of_gravity.html` (the scalar elastic-metric law; weak-field gravity closed scalar-only, including the factor-2 light bending, ray-traced 2.000 — the earlier tensor-response expectation is retracted there) and `WARP_DRIVE_MATH_AUDIT.md` (symbol table, two-regime conservation audit, the polarizable-vacuum floor $8\pi G/c^4$, and the closed-form dial $P_w = (S_{coh}{\cdot}f_{cons}){\cdot}\kappa{\cdot}½\varepsilon_r\varepsilon_0E^2$). Root ontology: `shape_of_the_universe.html` . Per-vertex bound: `G_VERTEX_DERIVATION.md` ($g \le 1/\sqrt{3}$). Substrate aggregate: `CHI_NU_AS_VERTEX_SHARING_STRENGTH.md` ($x_{sub} = \Sigma\, g_v\, \psi^*\psi$). Force law: `CHI_NU_FORCE_LAW.md` . Engineering-narrative companion: `warp_drive.html` (read through the audit). The architecture this paper specialises: `how_to_build_a_ufo.html` (§8 optimal substrate, §10 saucer morphology). The ground-effect prototype: `diy_hoverbike_paper.html` . Reproducible calculators: `hover/disk_material_ladder.py` , `hover/cymbal_chi_nu_force.py` .

§1 What this paper is

How to Build a UFO establishes the architecture: a disk-shaped coherent body that biases how the spacetime lattice settles in its neighbourhood, producing a net force its own mass-energy bookkeeping does not predict (the x_v / vertex-sharing channel). *The DIY Hoverbike* is the ground-effect prototype. This paper takes one question from §8 of the UFO paper and answers it in depth:

> *Of the materials we can actually obtain, which build a disk that reaches high x_v force — and why?*

This is an engineering/materials paper — its one enduring job is **substrate selection**. The mechanism itself is argued elsewhere: `the_shape_of_gravity.html` for what is actually derived (the scalar elastic-metric law, weak field closed) and `WARP_DRIVE_MATH_AUDIT.md` for the honest force-law bookkeeping. This paper does not claim a force; it takes the force law, isolates the *levers* the material sets — the four levers of §3 (energy ceiling, coherence, consonance, and coupling) — and ranks the candidate substrates against them. Every quantity is tagged DERIVED, ESTIMATE, or OPEN (bench-measured). The numbers are reproducible from `hover/disk_material_ladder.py`.

> **How to read the tags (and where the honesty lives).** DERIVED — follows from the force law or the geometry (the *form* $F = -\nabla E_{int}$, the bound $g \le 1/\sqrt{3}$). ESTIMATE — a heuristic *proxy*, a prior not a wall (notably $\boldsymbol{\eta \approx v_{phase}/c}$, §4). OPEN — a number only the bench fixes (notably the magnitude $\mathbf{C_{sub} \cdot |x_{sub}|^2}$ — earlier drafts wrote x_v. To be precise about the nulls: the bounds on it come from **published literature nulls** — Tajmar 2004, Bartlett & Tajmar 2021, Talley 1991, used as upper bounds in §15 — **the project's own bench has not yet measured any substrate**). BENCH — a build-and-measure step. No quantity in this paper is a demonstrated force; the paper ranks candidates to test, and the directional derivative $\partial|x_v|^2/\partial X$ (not η alone) is what a symmetric high-η disk would still need to be non-zero to lift at all (§3, §14.4).

§2 Why the lift is a materials problem, not an acoustics problem

A natural first instinct is to lift the disk acoustically — hold a trapped high-pressure region beneath it. That works *in ground effect* and only there. Steady hover requires a steady downward flux of momentum through any surface around the craft (Newton's third law). A pressure cushion routes that momentum into the *ground*, which is why it is cheap — but remove the ground and a genuinely-trapped, static overpressure exerts zero net force (internal forces cancel; you cannot lift yourself by pressurising a tank you carry), while a leaky one is just "throw air down" in disguise, with a power floor of tens of kW (a coherent jet) to ~500 kW (radiated sound) for a person-scale craft.

> **Derived · Free-altitude hover needs a non-ground momentum sink — and the conservation bookkeeping splits into two regimes** (`WARP_DRIVE_MATH_AUDIT.md` §2, the load-bearing honesty fix). **Regime (a)** — a gradient force on a *separate* object (levitation, tweezing, a cushion pressing the ground): textbook, can be large, not P/c-bounded. **Regime (b)** — net *self-propulsion* of the body's own centre of mass: a static conservative potential the body carries cannot do it ($\oint F \cdot dX = 0$, the bootstrap fallacy); it requires momentum handed to the lattice background, gated by $\langle F \rangle \lesssim \langle P_{coupled} \rangle / v_L$ with the lattice-mode speed v_L open, or a non-conservative time-asymmetric drive cycle (still v_L-gated). **Hover against Earth's field is the conservation-safe static-equilibrium case**: a static coupling force balancing weight in an external field does no work and ejects nothing — it is the near end, not the exotic end, of the ladder. A pressure cushion's reaction momentum terminates on the Earth, so cushion lift is intrinsically ground-effect; beyond it the only altitude-independent momentum sink in the framework is the lattice background itself (UFO §4). That sink *is* the substrate-coupling (x_v) channel. Hence a free-flying disk lifts by that channel, and how much force it makes is set by how well its material couples to the lattice.

So the disk's lift is a property of its *substrate*, not its aerodynamics. The rest of this paper is about choosing that substrate.

§3 The force law and the three things a material sets

From `CHI_NU_FORCE_LAW.md` (audited symbol-by-symbol in `WARP_DRIVE_MATH_AUDIT.md` §0–§1), the substrate-coupling force is an energy-gradient law. The coupling object is the substrate mode-overlap $x_{sub} = \Sigma\, g_v\, \psi^*\psi$ (OPEN — bench; earlier drafts, and the receipts below, write it x_v — same object, older name):

$$F \;=\; -\nabla E_{\text{int}} \;=\; \left(C_{\text{sub}}\,|\chi_\nu|^2\right) \;\times\; \left(\frac{U_{\text{drive}}\,\kappa}{L_{\text{grad}}}\right)$$

The second factor is the **known hardware lever**: stored coherent drive energy U_{drive}, drive-asymmetry directionality $\kappa \in [0,1]$ (the drive pattern is the force direction), and the gradient length L_{grad} ≈ the disk radius. The first factor is what the **material** sets. Three sub-quantities:

Factor	What it is	Status
$g_{max} = 1/\sqrt{3} \approx 0.577$	per-vertex geometric ceiling on coupling	DERIVED (G_VERTEX_DERIVATION.md)
$\eta \in [0,1]$	mode-match efficiency; $\lvert\chi_\nu\rvert = \eta\, g_{\max}$	naive value ESTIMATE; true value OPEN
U_{drive} ceiling	most coherent energy the body can hold	mechanical ceiling DERIVED; C_{sub} OPEN

g_{max} is fixed by geometry. What the material controls is the rest of the coupling, and it splits into **four levers** — the same four that `warp_drive.html` §13 / `WARP_DRIVE_MATH_AUDIT.md` §10 show are the four factors of the predicted thrust:

Lever	Material lever	Status
Energy ceiling (U_{drive})	specific resilience $\sigma^2/(E\rho)$ (§5)	DERIVED material lever
Consonance (f_{cons})	geometric match to $P = \varphi + \alpha\sqrt{\pi}$ — the lattice-match substrate (§13); ceiling $e^{-P/12} \approx 0.873$	sound logic; fidelity OPEN
Coherence (S_{coh})	Q and the coherent fraction (§5b)	DERIVED mechanism

Lever	Material lever	Status
Coupling (η)	mode-overlap η (§4)	OPEN — the A/B-fork lever (§4)

Provenance note. Earlier drafts headlined these four levers as Steiner's four ethers (Warmth / Chemical-tone / Life / Light, via `warp_drive.html` §13's four-ether decomposition). The correspondence is kept as provenance only — the technical lever names above are the load-bearing ones; nothing in the ranking depends on the ether reading.

In the audit's closed form the dial reads $\mathbf{P_w = (S_{coh}\cdot f_{cons})\cdot\kappa\cdot\frac{1}{2}\varepsilon_r\varepsilon_0 E^2}$ (`WARP_DRIVE_MATH_AUDIT.md` §10): the material's whole job is the coupling product $S_{coh}\cdot f_{cons}$, dialed between the **bare-vacuum floor** — the Einstein constant $8\pi G/c^4 \approx 2\times10^{-43}$, which is what the polarizable-vacuum correspondence (audit §8) says an *incoherent* body gets — and the coherent ceiling $f_{cons} \le e^{-P/12} \approx 0.873$. That the coherent enhancement actually climbs a real body off the floor is **the single open bet** of the programme (OPEN — bench); this paper ranks the materials that would climb furthest if it holds.

The thrust is the four levers' *product*, so a hover disk's job is to maximize all four. The correction this revision makes is to their *ranking*: three of them — energy ceiling, consonance, coherence — are sound material levers, while only the *frequency-scaling* of the coupling η is genuinely open (the A/B fork of §4). So the coupling η is **not the dominant lever — it is the *uncertain* one**, and the order below leads with it because it is the one the bench must settle, not because it outweighs the others.

§4 The coupling lever — the $\eta \approx v_{phase}/c$ proxy, and the open A/B fork

The substrate aggregate (CHI_NU_AS_VERTEX_SHARING_STRENGTH.md) is the mode-overlap of the substrate's coherent mode with the lattice mode at the drive frequency, $\eta = \lambda_{sub}/\lambda_{lat}$. Since $\lambda_{lat} = c/f$ and $\lambda_{sub}\cdot f = v_{phase}$ (the substrate mode's phase velocity), a first-pass overlap estimate collapses to a single ratio — a *heuristic proxy, not a derived law*:

$$\boxed{\eta \approx \frac{\lambda_{sub}}{\lambda_{lat}} = \frac{v_{phase}}{c}}$$

> **Heuristic (a prior, not a wall).** To first order η tracks the substrate's coherent-mode phase velocity over *c*: a vibration whose phase fronts move near light-speed overlaps the lattice mode coherently across the body; a slow vibration oscillates many times under one lattice wavefront

> and the overlap cancels. The whole "wavelength-mismatch suppression" the bench papers warn about is this ratio. But the *derived* object is the x_v overlap integral (CHI_NU_AS_VERTEX_SHARING_STRENGTH.md); this v_{phase}/c form is only its leading-order proxy, and the true η is OPEN (bench-measured).

This single ratio spans about six orders of magnitude depending on what physically carries the coherent vibration:

Mode (what carries the coherent vibration)	v_{phase}	η ≈ v/c
audio flexural (bronze bowl/cymbal, ~300 Hz)	~50 m/s	1.7×10^{-7}
bulk acoustic, steel/Ti ($\sqrt{E/\rho}$ ~ 5 km/s)	5.0×10^{3}	1.7×10^{-5}
bulk acoustic, diamond (~17 km/s)	1.7×10^{4}	5.8×10^{-5}
THz electromagnon / polariton (~0.1c)	3.0×10^{7}	1.0×10^{-1}
optical in n = 3.5 (Si photonic, c/n)	8.6×10^{7}	2.9×10^{-1}
surface plasmon (~0.5c)	1.5×10^{8}	5.0×10^{-1}

Because $|x_v|^2$ scales as η^2, *if* the v_{phase}/c proxy holds (**hypothesis A**, `warp_drive.html` §4), moving the coherent mode from mechanical to electromagnetic would buy **~10^8–10^{12} in coupling power** — which would make EM drive the single biggest material decision. **But that is exactly the open fork. Hypothesis B** — that the coupling is set by the *stored field*, frequency-independent — is equally live, and Buhler's *DC* vacuum result (§15) points at it. So "drive electromagnetic" is the *recommendation under hypothesis A*, not a settled lever; the §7 frequency sweep decides which fork holds. The other three levers — energy ceiling, consonance, coherence — hold either way.

4.1 The four open multipliers (UFO §8 predicates)

On top of the naive v_{phase}/c value sit multipliers that the framework predicts but only the bench can size. Each is a way to make the overlap add constructively beyond the bare velocity ratio:

- **P1 — icosahedral symmetry match.** A substrate whose own structural symmetry is the lattice's primitive (icosahedral) cell has an overlap integral that accumulates rather than cancels across the body.

- **P2 — a coupling channel that ties spin to the lattice.** Multiferroic order gives two channels in concert — but its *rigorous* form is **spin–orbit coupling** (the operator $\lambda L \cdot S$), the only interaction that mechanically fuses spin to real-space orbital/lattice motion. SOC strength λ is the natural figure of merit; the 5d iridate $J_{eff}=1/2$ state ($\lambda \approx 0.5$ eV) maxes it. See §12.
- **P3 — phase-coherent condensate at the boundary.** A Cooper-pair / Bose / optical condensate preserves coupling phase across the body; a thermal boundary randomises it. Its one *room-temperature* realisation is the **magnon (spin-wave) Bose-Einstein condensate** — a body-spanning, single-phase spin condensate (§12).
- **P4 — drive frequency near the lattice mode.** This is the v_{phase}/c term itself — the reason THz-to-optical drive is the framework's prescription *under hypothesis A*; whether frequency is a lever at all is the open A/B fork (§4), which the §7 sweep decides.

> **Open — bench measures** · The sizes of P1–P3 are the **open numbers**. The naive $\eta = v_{phase}/c$ is a lower-bound-flavoured prior that assumes none of them. A lattice-matched, multiferroic, condensate-boundary body could have η well above its bare velocity ratio — which is exactly why the substrate ladder is an experimental program and not a closed calculation. *(Anti-pessimism guardrail: a low naive η is a prior, not a wall.)*

4.2 The light-consuming corollary — a cheap first-pass filter

If coupling to the lattice goes as the substrate's electromagnetic mode-overlap, then a material that couples optical-frequency EM into a coherent mode is, in the visible band, *dark* — the same overlap that biases the lattice also absorbs light. Absorptance should track $|x_v(\omega)|^2$ in the coupling band. (This is the optical face of the coupling developed in `how_to_build_a_ufo.html` §11.5, and reconciles Lazar's paired report of a craft that "consumes light *and* bends gravity" — two frequencies of one high-frequency mode-overlap, not two marvels.)

> **Implementation consequence** · **"Is it anomalously, broadbandly black at its coupling band?" becomes a cheap screen.** A good x_v optical coupler must be unusually light-consuming; if a candidate is optically ordinary, the high-η reading is falsified. Strikingly, this filter points straight at the §7 top picks — **graphene's opacity is literally the fine-structure constant** ($\pi\alpha = 2.3\%$ per layer, broadband, gate-tunable; Nair 2008), **CNT forests are the blackest known material** (>99.99%), and the Pt-group "blacks" (including iridium) sit just behind. The honest null to keep in mind: a *passive* super-absorber (a Vantablack-class coating) is black without any x_v coupling, so

blackness is a necessary screen, not a sufficient proof — it filters candidates in, it does not confirm them.

§5 The energy-ceiling lever — specific resilience $\sigma^2/(E\rho)$

U_{drive} is *stored* coherent energy, not dissipated power — so cavity Q raises efficiency (you reach a given energy at lower power) but not the ceiling. For a body that stores the energy *mechanically*, the ceiling is set by the stress at which it fractures or fatigues. Integrating the elastic energy density $\sigma^2/(2E)$ over the volume that participates near peak stress (a flexural mode reaches the working stress only at the surface antinodes) gives

$$U_{cap} = \frac{\sigma_{work}^2 V}{12\,E} \quad\Longrightarrow\quad \frac{U_{cap}}{\text{mass}} = \frac{\sigma_{work}^2}{12\,E\rho}$$

So the relevant material figure of merit is the **specific resilience** $R = \sigma_{work}^2/(E\rho)$ — the elastic energy a material stores per unit weight. It sets a clean, *size-independent* ceiling on x_v self-lift at perfect coupling:

$$\left(\frac{F}{W}\right)_{perfect} = \frac{\sigma_{work}^2\,\kappa\,g_{max}^2}{12\,E\rho\,g\,L_{grad}} \propto R$$

Derived · The body's mass cancels. A bigger disk is heavier in exact proportion, leaving only a weak $1/L_{grad}$ ($\propto$ 1/radius) residual. So whether a material can self-lift via x_v — even at perfect coupling — is a property of the *material*, not the size.

Ranking candidate structural bodies (sustained / fatigue-class working stress; self-lift quoted for a 1 m-radius disk, L_{grad} = 0.5 m):

Structural material	σ_{work} (MPa)	c_{sound} (km/s)	R (J/kg)	self-lift @ perfect
Single-crystal diamond	2000	17.3	1084	3.07 ✓
CFRP (uniaxial)	600	9.2	1667	4.72 ✓
Maraging steel (18Ni-300)	1200	4.9	947	2.68 ✓
Ti-6Al-4V	500	5.1	495	1.40 ✓
Silicon carbide (4H)	500	11.8	173	0.49 ✗

Structural material	σ_{work} (MPa)	c_{sound} (km/s)	R (J/kg)	self-lift @ perfect
Beryllium	240	12.5	108	0.31 ✗
Sapphire (cryo BAW)	400	10.0	101	0.28 ✗
Spring / high-strength steel	700	5.0	312	0.88 ✗
Stainless 304	200	4.9	26	0.07 ✗
Fused quartz (crystal bowl)	50	5.7	16	0.04 ✗
Bell bronze (singing bowl)	100	3.4	11	0.03 ✗

Bronze and quartz — the singing-bowl materials — physically *cannot* self-lift via x_v even with perfect coupling: their specific resilience is too low. Diamond, CFRP, maraging steel, and titanium clear F/W > 1. (Spring steel sits just under at 1 m radius and clears it for a smaller disk.)

§5b The coherence lever (S_{coh}) — the isotopic knob on Q and the coherent fraction

The coupling (§4) and energy-ceiling (§5) levers set what a material could do at *perfect* coupling and at its energy ceiling. This third lever sets how much of that you realise: **coherence**. It enters twice. First, $U_{drive} = Q \cdot P_{in}/\omega$, so at fixed input power a higher quality factor Q stores more coherent energy (§5). Second, the substrate aggregate is built only from the *phase-coherent* fraction of the body — thermalised, decoherent matter contributes incoherently and averages away (`CHI_NU_AS_VERTEX_SHARING_STRENGTH.md`). So longer coherence raises both U_{drive} (via Q) and the effective $|x_v|$ (via the coherent fraction). It multiplies the force in the realistic, power-limited regime; it does not raise the mechanical ceiling of §5, but it sets how much of that ceiling is actually coherent.

> **Derived · The cleanest way to raise coherence is isotopic.** The dominant decoherence channel in a solid is the nuclear-spin bath, removed by *isotopic purification* — enriching the spin-zero or spin-depleted isotope. The mainstream, UFO-independent evidence is unambiguous: **^{28}Si enriched to 99.7%** (depleting spin-½ ^{29}Si) extends spin coherence to **~10 ms**; **spin-0 ^{12}C diamond** is the canonical low-decoherence host for NV centres. Isotopic engineering is a real, makeable materials technique (isotopically-enriched CVD diamond and ^{28}Si are commercial), not an exotic claim.

Derived · **It strengthens the diamond core specifically.** Single-crystal diamond already wins both ceiling levers (§5–6): structural champion (R = 1084) *and* high-index optical substrate. **^{12}C-enriched diamond adds the third** — spin-0 host, longest demonstrated solid-state coherence. So the diamond core is structural, optical, *and* coherent in one body. Apply the same knob to any candidate: an isotopically-pure variant of the chosen substrate is a free multiplier on its coherent fraction.

Why this lever is here, and why it costs the paper nothing in honesty. The independent reverse-engineering efforts on purported recovered materials (Nolan & Vallée, *Prog. Aerospace Sci.* 2022; the AARO/Oak Ridge analysis, 2024) converged on *isotope ratios* as the signature to test — and *every analysed sample came back terrestrial* (the much-quoted “30%-off magnesium” was a single-lab artefact the primary data contradict). The useful, makeable residue of that work is not any sample but the *technique*: isotopic engineering for coherence, citable to the qubit literature and independent of any UFO claim. We adopt the technique and discard the samples.

§6 Coupling × energy ceiling — the naive self-lift estimate

The naive x_v self-lift is the product of the coupling and the energy ceiling. Using $|x_v|^2 = (\eta \cdot g_{max})^2$ it reduces to a clean form, *self-lift* = $(F/W)_{perfect} \times \eta^2$:

Body / drive mode	η_{naive}	$(F/W)_{perfect}$	naive self-lift F/W
Bell bronze, audio flexural	1.7×10^{-7}	0.03	9×10^{-16}
Ti-6Al-4V, hypersonic acoustic	1.7×10^{-5}	1.40	4×10^{-10}
Diamond, hypersonic acoustic	5.8×10^{-5}	3.07	1×10^{-8}
Diamond, optical mode	4.1×10^{-1}	3.07	**0.52**
SiC, optical mode	3.8×10^{-1}	0.49	**7×10^{-2}**

Doctrine sentence · *A bronze singing bowl driven at audio frequency self-lifts about 10^{-15} of its own weight. A diamond disk driven in its optical mode self-lifts about half its own weight — on the same naive estimate, before any of the open multipliers, and with the magnitude $C_{sub} \cdot |x_v|^2$ still unmeasured (bounded only by the published literature nulls of §15; our own bench has measured no substrate yet). If the $\eta \approx v_{phase}/c$ proxy holds (hypothesis A, §4), material and drive-mode choice span ~13–14 orders of magnitude of the x_v lift gap; the energy-ceiling half of that span is solid, while*

> *whether the drive-mode (frequency) half is real is the open A/B fork. Either way the substrate is the machine — and the convergent machine is not one crystal but the* ***layered stack of §8****; the diamond-optical single body is its first tier, not its endpoint. In the* [redacted] *§13 closed-form thrust, a hover disk needs only $\sim 10^{-6}$ of full coherence to self-lift, so it is the near-term end of the one dial that runs to the FTL bubble.*

The remaining factors are C_{sub} (the substrate normalization, ≤ 1) and the P1–P3 multipliers (which act *above* the naive η). The disk does not need a coupling miracle; it needs the right body driven in the right mode, and then the bench decides whether the open factors land it above or below unity. (The optical figure conservatively uses the mechanical energy ceiling; wide-bandgap diamond's optical-damage ceiling is plausibly higher.)

§7 The best options, in detail

Each class below is rated on the coupling *η* and the energy ceiling *R*, plus which open predicates it carries — P1 (consonance / symmetry match), P2 (coupling channel), P3 (coherence). That is the four levers of §3 spread across the columns. No single material maxes everything; §8 assembles them.

The scoreboard — which material does which job

Class	Coupling η	Ceiling R (J/kg)	Open predicates	Role on the disk
Graphene / CNT	η uncertain 0.003–0.2*	7.5×10^{6} (champion)	P1 (12-fold) + P3 (MATBG)	resilience core; η-conditional skin
Diamond	optical 0.41	1084	—	core + skin (both)
SiC	optical 0.38	173	—	scalable core + skin
Ti / maraging / CFRP	acoustic $\sim10^{-5}$	495–1667	—	structural core
Quasicrystal (Al-Pd-Mn)	—	low	P1 symmetry	symmetry layer
$BiFeO_3$	THz 0.1	low	P2 + P4	coupling film

Class	Coupling η	Ceiling R (J/kg)	Open predicates	Role on the disk
Iridate (Sr_2IrO_4)	THz strain-tunable	low	P2 (giant ME) + P4 + nuclear	coupling oxide
Superconductor (Bi / cuprate)	—	low	P3 condensate	coherent boundary
Photonic / plasmonic	0.3–0.5	n/a (optical)	—	active η surface
Iridium *metal*	acoustic ~10^{-5}	336 (brittle)	—	poor — use the oxide

* Graphene's η is genuinely uncertain (see §7.8): the famous v_F = 0.003c is a *transport* velocity (a category error to use as η); bare plasmons are slow; the optimistic 0.1–0.3 is an hBN-hybrid *group* velocity. Its resilience is so high that even the pessimistic η self-lifts ~0.2× its weight.

7.1 Single-crystal diamond — the convergent single body

Diamond is the one material that wins *both* levers outright. It is the structural champion (highest sound speed of any material, 17 km/s; fracture strength of CVD diamond ~2 GPa; specific resilience 1084 J/kg, self-lift 3.07), and simultaneously a high-index optical substrate (n = 2.42 → η ≈ 0.41 in its optical mode). It has the highest known thermal conductivity (it sheds drive heat), a wide bandgap (high optical-damage threshold, so the optical energy ceiling is comfortable), and lab-grown CVD plates are now manufacturable at centimetre scale and growing. A diamond disk reaches the electromagnetic-η regime *without a separate active layer* and holds the energy in the same body. **If you build one disk from one material, build it from diamond.**

7.2 Silicon carbide — the scalable second

SiC (4H polytype) is diamond's cheaper, larger-wafer cousin: high sound speed (11.8 km/s), optical substrate (n = 2.6 → η ≈ 0.38), mature wafer and CVD manufacturing at 150–200 mm. Its specific resilience is lower (173 J/kg, self-lift 0.49 at 1 m), so a SiC disk is a strong η body that may need a high-R backing layer to clear self-lift. The natural prototype substrate before diamond is affordable at disk scale.

7.3 Titanium, maraging steel, CFRP — the structural cores

These win the energy ceiling but only reach acoustic η on their own (~10^{-5}). Their role is the *structural core* that holds the coherent energy beneath an active η-skin. Ti-6Al-4V (R 495, self-lift 1.40) is the aerospace default: weldable, available, fatigue-characterised. Maraging steel (R 947, self-lift 2.68) trades density for a higher ceiling. CFRP has the highest specific resilience of all (1667 J/kg, self-lift 4.72) — but it is anisotropic and a fibre/matrix composite, so its coherent-mode Q and phase coherence are poor; treat it as a mass-efficient backing, not a coherent substrate.

7.4 Icosahedral quasicrystals — the symmetry layer (P1)

Al-Pd-Mn and Al-Cu-Fe icosahedral quasicrystals (Shechtman 1984; Tsai 1987) are the only substrate class whose own diffraction symmetry *is* the lattice's icosahedral primitive cell. That is predicate P1 in material form: a potentially large, open multiplier on η from symmetry-matched overlap. They are brittle (low R), so they serve as a thin *symmetry layer* on a structural core, not as the load body. This is the substrate rung that tests whether the symmetry-match multiplier is real and large.

7.4b Bismuth — rung 2 of the bench ladder (and the Bi/Mg layered waveguide)

The bench substrate ladder runs **pot-lid steel → bismuth → $BiFeO_3$ → quasicrystal** (UFO paper; `UFO_LANE_MIGRATION_NOTE.md`), and bismuth is its rung 2 — the first substrate past the null-hypothesis steel, and the cheapest one that stacks several coupling-relevant anomalies in a single elemental solid:

- **Strongest simple diamagnet.** Elemental bismuth has the largest diamagnetic susceptibility of any ordinary (non-superconducting) material — it is the natural bridge substrate between the §11 diamagnetic control and any claimed coupling beyond it.
- **Semimetal with very low carrier density and long mean free path.** Carriers ~10^{-5} per atom with mean free paths reaching millimetres in pure crystals — an unusually clean, quiet electronic background for a coherent mode, and the setting of its unexplained 0.53 mK superconductivity (Prakash et al. 2017, §7.6).
- **Heavy atom, strong spin–orbit coupling.** Bismuth is the parent of the Bi_2Se_3 / Bi_2Te_3 topological-insulator family (and of higher-order topology in elemental Bi; Schindler 2018) — and of **$BiFeO_3$ itself**, the ladder's rung 3 (§7.5). Measuring rung 2 before rung 3 separates "what the Bi atom's SOC buys" from "what the multiferroic order adds."

> Open — bench measures · **The Bi/Mg layered material — honest register (candidate / spec, not receipt).** The famous micron-layered bismuth–magnesium artifact of UAP lore was

> assessed in AARO's *Historical Record Report* (2024, with the Oak Ridge analysis): **ordinary and terrestrial**. It therefore carries **no evidentiary weight** in this paper and is marked so. What survives is pure engineering inspiration, detached from any sample's provenance: alternating light-metal/semimetal micron layers as a **THz waveguide** is a real, testable metamaterial hypothesis. Its honest slot in this paper's architecture is **drive-plumbing** — a THz distribution/waveguide layer that routes the drive across the disk — **not** the coherent-storage layer: metals cannot store an E-field internally (§14's conductor-shorts point), the stack has no icosahedral order (no P1), and its coherence properties are unknown. It is fabricable today by sputtering alternating Bi/Mg layers at controlled sub-micron periodicity, which makes it a **cheap bench rung**: build the clean version, measure its THz transmission and any coupling residual, and let the numbers speak. [CANDIDATE/SPEC] throughout — nothing here is a receipt.

7.5 Multiferroics ($BiFeO_3$) — the coupling film (P2 + P4)

Bismuth ferrite carries simultaneous ferroelectric and antiferromagnetic order (two coupling channels, P2) and supports THz electromagnon modes (P4, η into the ~0.1 regime). Cycloid-suppressed epitaxial thin films reach the strongest room-temperature magnetoelectric coupling known (~3 V/cm·Oe; Catalan & Scott 2009). Grown as a thin film on the structural/symmetry core, $BiFeO_3$ is the *active coupling layer* that drives the body into the electromagnetic-η regime through a broken-symmetry order parameter rather than through bulk motion.

7.6 Superconducting condensates — the coherent boundary (P3)

A phase-coherent condensate at the boundary preserves coupling phase across the whole body (P3) — the lineage of the Ning-Li and Tajmar rotating-superconductor experiments (UFO §9). Cuprate films give P3 at 77 K; MgB_2 at ~39 K. Bismuth is the standout: its 0.53 mK superconductivity is six orders below any BCS prediction and has no accepted mechanism (Prakash et al. 2017), so it may escape the s-wave Canterbury-2008 bound that constrains conventional superconductors. The condensate is a surface boundary layer, not the structural body.

7.7 Photonic and plasmonic skins — the η extreme

The highest η of any buildable substrate. Silicon photonic ($n \approx 3.48$, $\eta \approx 0.29$) is the optical-chamber substrate class; a metallic plasmonic skin reaches $v_{phase} \sim 0.5c$ ($\eta \approx 0.5$). Energy is stored in the optical/plasmonic field (high damage ceiling), not as mechanical strain, so the energy ceiling is comfortable. These are not structural bodies — they are the η reference point and the natural "active surface" of a diamond/SiC disk.

7.8 Graphene — the resilience champion (energy ceiling by ~7000×)

Graphene is the highest-specific-resilience material known: $\sigma \approx 130$ GPa, $E \approx 1$ TPa, $\rho \approx 2267$ kg/m³ give $R \approx 7.5\times10^6$ J/kg — about **7000× diamond** (single-wall CNTs match it at the theoretical limit; real CNT fibre ~4× lower). Its energy ceiling is so high that the self-lift question inverts: at perfect coupling it would lift ~7000× its weight, so the *only* thing that matters is how much η it can reach. The honest range (self-lift = $(F/W)_{perfect} \times \eta^2$):

η source	η	naive self-lift F/W
bare confined plasmon (phase vel ~c/300) — pessimistic	0.003	~0.2
hBN-hybrid polariton (group vel ~0.2c) — optimistic	0.2	~840
perfect coupling ($\eta \to g_{max}$)	0.577	~7000

> **Open — bench measures** · Graphene's η is the open, contested number. The $v_F = 0.003c$ Dirac velocity is a *quasiparticle transport* speed, not a coherent mode's *phase* velocity — do not credit graphene with EM-band η on its strength. And the extreme plasmon confinement that makes graphene exciting also makes those plasmons *slow* (confinement and high η pull opposite ways). The best honest η is the graphene–hBN hybrid plasmon-phonon polariton (~0.1–0.3, and that is a group velocity). But because R is ~7000× diamond, **even the pessimistic slow-plasmon η self-lifts ~0.2× weight on the naive estimate** — graphene is robust to η uncertainty in a way diamond is not.

Three further framework hits make carbon hard to ignore: (i) **30°-twisted bilayer graphene is a genuine dodecagonal (12-fold) quasicrystal** — predicate P1, and a direct match to the project's 12-port / icosahedral geometry; (ii) **magic-angle twisted bilayer graphene superconducts** ($T_c \approx 1.7$ K, geometry-set) — predicate P3, a condensate switched by a twist angle, conceptually resonant with vertex-sharing; (iii) graphene's **opacity is the fine-structure constant** (§4.2). Caveats: graphene has essentially no native P2 (weak spin–orbit, non-magnetic — needs proximity engineering), MATBG's T_c is cryogenic and fabrication-fragile, and graphene *aerogel/foam is a trap* — going ultralight collapses σ faster than ρ, dropping R ~100× below monolayer. Rank by $\sigma^2/(E\rho)$, never by density.

7.8b Carbon nanotubes — the light-consuming aperture and the spin handle

CNTs share graphene's single-tube resilience (~7×10^6 J/kg) but lose it in any buildable form (fibre/yarn/buckypaper fall to 0.01–1.2×10^6 J/kg as tubes slip on their van-der-Waals contacts) — so CNTs are *not* the Lever-2 win. They earn their place on two CNT-specific axes graphene cannot match:

- **The light-consuming aperture (decisive).** A vertically-aligned CNT *forest* is the blackest known material — Vantablack ~99.965%, the MIT 2019 record ≥99.995%, broadband UV→THz and angle-independent — because it *multiplies* intrinsic 1D broadband absorption by geometric light-trapping. That product is the physical realisation of the §4.2 filter: the forest forces every entering mode to sample an enormous number of surface vertices before it can leave — structurally the same "coherent aggregation over many boundary vertices" as the x_v overlap sum. A flat graphene sheet has the intrinsic coupling but none of the geometric aggregation (2.3% vs 99.995%). So if a good optical coupler must be black, **the CNT forest is the substrate that operationalises the coupling aperture**, and graphene is not.
- **Curvature spin–orbit coupling (P2).** Wrapping graphene into a tube breaks its mirror symmetry and turns on first-order SOC — measured at 0.37–3.4 meV (a built-in ~29 T axial field), i.e. **10^3–$10^4\times$ flat graphene's ~1 μeV**, and geometry-tunable by diameter and chirality. CNTs also carry native 1D superconductivity (intrinsic at 0.4 nm, $T_c \approx 15$ K; plus rope and proximity SC) — a P3 channel graphene lacks natively — and a quantized Luttinger-liquid plasmon at $\eta \approx 0.008$–0.01 (carrier-density-independent, cleaner than a 2D plasmon). Chirality is also a selectable P1 knob.

> **Implementation consequence · Two honest caveats.** (1) The blackness→x_v inference is the framework's own postulate — a forest's blackness is fully explained by ordinary absorption into heat, with no x_v required; it is a necessary screen, not a proof. (2) The morphologies fight: the 99.995%-black *forest* is a fragile low-fill-fraction array, while a load-bearing disc wants dense *fibre* — you cannot get both the black aperture and the structural body from one piece. On the "Sport-Model black panels" lore: it is *not* documented — Lazar's craft was described as bare metallic, and the famous stealth blacks are *radar* (magnetic-loss) absorbers, a different mechanism from optical ultra-black. The CNT-aperture case rests on the laboratory blackest-material record alone, which is enough.

7.9 Iridates — the coupling oxides (it's the compound, not the metal)

Iridium-*the-metal* is a poor candidate: dense and brittle (intrinsic Frank-loop cleavage), specific resilience ≤ steel (R ≈ 336 J/kg at cold-worked strength), acoustic-only coupling, and superconducting only at 0.11 K. The framework-relevant physics lives entirely in the **iridates** (Ir-oxides), one of the richest playgrounds for exotic coherent modes:

- **Sr_2IrO_4** — a spin–orbit-coupled J_{eff}=1/2 Mott insulator (a near-perfect cuprate analogue) with a **giant magnetoelectric effect** (predicate P2, two coupling channels via an unconventional SOC mechanism).
- **THz collective modes** — dispersing magnons up to ~50 THz plus a spin–orbit exciton, intrinsically in the high-frequency band (P4), and **strongly strain-tunable** — i.e. the coupling mode is itself a handle that ties the coupling η to the energy ceiling.
- **A nuclear↔electronic channel** — ^{191}Ir was the original Mössbauer nuclide; ^{193}Ir nuclear-resonant scattering reads the magnetic state, and a giant magnetic isotope effect appears in an Ir quantum-spin-liquid. A genuine second coupling channel few elements offer.
- Honeycomb/pyrochlore iridates host Kitaev-spin-liquid and Weyl-semimetal states (P1-flavoured frustrated geometry), though all real iridates order magnetically rather than realising the ideal spin liquid.

Caveat: predicate P3 is *unfilled* for iridates (no confirmed iridate superconductor yet), and iridium is among the rarest, most expensive elements — a practical ceiling on any build. Use the oxide as a thin coupling layer, pair it with a separate condensate boundary for P3, and treat the ORMUS / "monatomic-iridium" claims as a *testable lead* rather than established physics — see §7.10.

7.10 Wildcard (speculation) — ORMUS / monatomic Pt-group

> **Open — bench measures** · **Listed for completeness, not relied upon.** The headline ORMUS claims — bulk room-temperature superconductivity, ~4/9 weight loss, arc-made permanent high-spin *nuclei* — are refuted: a room-temperature superconductor is trivially SQUID-detectable; the weight anomaly is a thermogravimetric buoyancy artifact; high-spin nuclei are picosecond accelerator states, not a storable powder; commercial "ORMUS" assays as magnesium hydroxide. The one *untested sliver* is real — a noble-metal atom stripped of bulk metallic bonding occupies an anomalous d-hole state with new magnetism, vindicated in spirit by thiol-capped gold nanoparticles showing room-temperature ferromagnetic hysteresis (Crespo, PRL 2004). That is a legitimate **P2 (d-hole spin) channel**, and no one has put a genuine Hudson-route Ir/Rh preparation through a SQUID + EPR battery and published it. The decisive test is one afternoon on a SQUID/VSM: it falsifies the superconductivity outright and detects any real moment to ~10^{-7} emu. Investigate the fringe by *measuring* it — do not build on it.

§8 The convergent disk — winning every lever by layering

No single material maxes every lever *and* carries all the predicates, so the high-end disk is a stack, structured exactly along the §6 morphology of the UFO paper (closed coherent shell, rim-segmented drive, top–bottom asymmetry, central cabin):

Layer	Job	Material
Structural core	holds the coherent energy (energy-ceiling lever, §5) + coherence (§5b)	diamond (best; **^{12}C-enriched** for coherence); else maraging steel / Ti / CFRP backing
Symmetry layer (P1)	lattice-cell symmetry match (consonance, §13)	Al-Pd-Mn icosahedral quasicrystal
Coupling film (P2, P4)	two channels; THz drive targets the EM-η regime (hypothesis A, §4)	epitaxial $BiFeO_3$
Condensate boundary (P3)	preserves coupling phase across the body	cuprate film (77 K); bismuth's own condensate physics is 0.53 mK (§7.6) — cryogenic, not a room-T P3
Drive plumbing	THz distribution / waveguide layer (routes the drive; stores nothing)	clean, lead-free **Bi/Mg layered superlattice** (§7.4b — candidate/spec register)
Drive	asymmetric, directional ($\kappa \to 1$)	rim-segmented THz-to-optical sources

> **Implementation consequence** · The single-body shortcut is a **diamond disk driven optically**: it carries the coupling η (optical $\eta \approx 0.41$, hypothesis A) and the energy ceiling ($R = 1084$, self-lift 3.07) in one material, no active layer required. The full multilayer stack is what you build when you want the P1–P3 multipliers *on top* of that — i.e. when the bench has shown they are large enough to be worth the fabrication cost. Diamond first; predicates second.

> **The Bi/Mg layered rung: designed, not industrial — and drive-plumbing, not storage.** The full register is §7.4b; the short form: the famous layered Bi/Mg artifact was assessed by AARO (2024 report, with the Oak Ridge analysis) as **ordinary and terrestrial** — a *lead-contaminated* Bi/Mg-Zn slag is the natural by-product of the Betterton–Kroll lead-refining process, and every purported recovered "metamaterial" analysed to date (the Art's-Parts samples; the AARO/ORNL specimen) resolved into exactly that. It carries no evidentiary weight. The engineered target that survives as inspiration is a **clean, lead-free Bi/Mg superlattice with controlled sub-micron**

> **periodicity** — a testable THz-waveguide metamaterial, sputterable today, whose honest slot in this stack is the *drive-plumbing row* (it routes the THz drive; it cannot be the coherent-storage layer — metals hold no internal E-field, and it has neither icosahedral order nor known coherence). Build-and-measure target with no reliance on any sample's provenance. [CANDIDATE/SPEC]

Steering, cabin, and drive electronics follow the saucer morphology unchanged (UFO §10): per-segment rim drive gives directional thrust by amplitude/phase asymmetry; the top–bottom drive asymmetry makes the disk a lift platform rather than a sphere; the central thickening houses crew, power, and the THz/optical drive sources.

§9 Build tiers

Tier	Body	Drive	What it answers	Order of cost
0 — demonstrator	Ti or steel disk on a pendulum	audio / ultrasonic	extends the pot-lid bench to a disk; bounds acoustic-η at the disk geometry	$ (bench)
1 — research	SiC then diamond disk	optical	measures EM-regime η; first test of the §6 headline at a torsion-balance readout	$$$
2 — aspirational	full multilayer stack	THz + optical, rim-segmented	turns on the P1–P3 multipliers; the §8 disk	$

The ladder is deliberately the same shape as the bench substrate ladder in the UFO and pot-lid papers (pot-lid steel → **bismuth, §7.4b** → $BiFeO_3$ → quasicrystal): each tier produces a clean η measurement at one substrate + geometry, and the program is the ranking of those η values — not a single go/no-go. A bismuth slab (and the sputtered Bi/Mg layered waveguide of §7.4b) slots between tiers 0 and 1 as a cheap elemental rung.

Two **pre-bench screens cost almost nothing** and should run first. (i) *The reflectance/blackness screen* (§4.2): measure broadband absorptance of each candidate on a spectrometer — a good optical x_v coupler must be anomalously black at its coupling band, so this filters the §7 picks in (and any optically-ordinary candidate out) before any force apparatus is built. (ii) *The diamagnetic control* (§11): pyrolytic graphite over a NdFeB array levitates with no power and pins the conventional diamagnetic baseline, so a later force result in a field can be distinguished from ordinary diamagnetism.

§10 What is derived, estimated, and open

Element	Status
$g_{max} = 1/\sqrt{3}$ per-vertex ceiling	DERIVED
Force law $F = (C_{sub}\|x_v\|^2)(U_{drive}\kappa/L_{grad})$	DERIVED form (energy gradient)
$\eta \approx v_{phase}/c$ (coupling)	ESTIMATE (naive overlap; the open **A/B fork**, warp §4 — a prior, not a wall; *not* the dominant lever)
Energy ceiling $U_{cap} = \sigma^2V/12E$; specific resilience R	DERIVED (mechanical)
Self-lift ceiling ∝ R (size-independent)	DERIVED
Coherence lever (Q, coherent fraction S_{coh}) via isotopic purity (§5b)	DERIVED mechanism; isotopic enrichment a known, makeable process
Consonance — geometric match to P (lattice-match substrate, §13); f_{cons} ceiling $e^{-P/12} \approx 0.873$	sound logic; the consonance FIDELITY is OPEN (bench)
Closed-form dial $P_w = (S_{coh} \cdot f_{cons}) \cdot \kappa \cdot \frac{1}{2} \varepsilon_r \varepsilon_0 E^2$ (audit §10)	DERIVED form — a falsifiable prediction, not a measurement
Bare-vacuum coupling floor $8\pi G/c^4 \approx 2\times10^{-43}$ (audit §8, PV correspondence)	DERIVED (Einstein constant — what an incoherent body gets)
Coherent enhancement above the PV floor — **the single open bet**	OPEN — bench; literature nulls (§15) bound it, our bench has not yet measured it
P1–P3 multipliers on η; C_{sub}	OPEN — the bench measures where a body sits on the dial

The honest one-line state: the levers and ceilings are derived; the naive η is a conservative prior; the coherence lever is a real, makeable multiplier; and the P1–P3 multipliers and C_{sub} are what the substrate ladder is built to measure. The material analysis tells you *where to build* (diamond / SiC, optical mode, **^{12}C-enriched** for coherence, clean Bi/Mg-Zn boundary, with the predicate layers as upside); it does not by itself certify a Newton of force. That certification is a bench receipt, per the schema the pot-lid and UFO papers share.

Doctrine sentence · *The hover disk's lift is its substrate — four levers (§3): the energy ceiling (specific resilience → go diamond), coherence (isotopic purity), and consonance (lattice-match) are sound material levers; the coupling's frequency-scaling ($\eta \approx v_{phase}/c$ → go electromagnetic) is the open A/B fork the §7 bench settles. Maximize the four and the lift gap closes to the span of the open coupling the bench reads. The predicted thrust and the coherence dial $P_w = (S_{coh} \cdot f_{cons}) \cdot \kappa \cdot \tfrac{1}{2} \varepsilon_r \varepsilon_0 E^2$ are* `WARP_DRIVE_MATH_AUDIT.md` *§10 /* `warp_drive.html` *§13 — a hover disk needs only ~10^{-6} of full coherence to lift.*

§11 Diamagnetic levitation — the conventional control (and the Meissner→P3 bridge)

One conventional, room-temperature, fully-demonstrated mechanism lifts ordinary matter — Geim's levitating frog (Berry & Geim 1997). All matter is weakly diamagnetic, so it is repelled from high field; the on-axis balance is

$$B \frac{dB}{dz} = -\frac{\mu_0 \rho g}{|\chi|}$$

It depends on the object's *density, not its mass* (frog, water and hazelnut float at the same setting) and cancels gravity atom-by-atom throughout the bulk — the cleanest existing uniform body force on a whole coherent object. **Pyrolytic graphite, the best room-temperature diamagnet, levitates over ordinary NdFeB with no power at all** (the cheap demonstrator — and it closes the carbon thread, being also the §7.8 specific-resilience champion).

> **Implementation consequence** · This is force-balance against a real magnetic body force — **not** anti-gravity and **not** the x_v channel; do not over-read it. It is load-bearing as the *control*: a disk that lifts in a strong field may simply be diamagnetic, so any x_v result must out-perform or be distinguished from it. And its perfect-diamagnet limit, $x = -1$, is the **Meissner effect** — a superconductor expelling the field — which is exactly predicate **P3**. So the superconducting-boundary layer of the §8 disk is simultaneously the strongest diamagnetic levitator *and* the x_v coherence predicate: the two pictures meet at the condensate.

§12 Spin — the rigorous content of P2, and the room-temperature P3

"Spin" appears in every candidate above, meaning several different things; pulled apart it sharpens two predicates and points at one genuinely additive material. The orienting fact: **spin–orbit coupling is the only known interaction that mechanically ties spin to real-space geometry** — the operator $\lambda L \cdot S$ entangles spin with lattice-locked orbital motion, so SOC strength λ is the figure of merit for P2. It climbs three orders across the §7 candidates: graphene ($\sim 1\ \mu$eV) → CNT curvature (0.4–3.4 meV, §7.8b) → 5d iridate (λ ≈ 0.5 eV, building the $J_{eff}=1/2$ state, §7.9). The carbon tension is built in: the low atomic number that gives ^{12}C its *long spin coherence* (§5b) is exactly what gives it *negligible SOC* — you cannot maximise both in one carbon.

Derived · The room-temperature P3 prize is the magnon BEC. Of every coherent mode in this paper, the magnon Bose–Einstein condensate in YIG is the only one that is at once (i) *room-temperature*, (ii) a genuine body-spanning single-wavefunction condensate (measured spin supercurrents, a Josephson analogue), (iii) natively spin-carrying (a phonon or plasmon is not), and (iv) GHz–THz drivable (P4). It is the cleanest realisation of "a phase-coherent condensate that holds coupling phase across the body" — and it exists today, at 300 K.

Open — bench measures · But it is not an η win, and the coupling is unmeasured. A magnon's phase velocity is ~km/s, so its $\eta \approx v_{phase}/c \approx 3\times10^{-5}$ is phonon-like (§4). Its case rests entirely on the other axis: it carries spin and stays coherent across the body at room temperature. So the spin channel and the η (electromagnetic-mode) channel are *different levers* a full disk would combine — a high-η optical/diamond aperture coupled to a room-temperature spin condensate — not one material. And every "spin couples to x_v" statement is unmeasured hypothesis: the materials physics (the λ ladder, the 300 K magnon BEC) is established; whether any of it feeds the x_v channel is what the bench must decide. Two clean tests fall out — **(1) the SOC ladder:** does the coupling scale with λ across graphene → CNT → iridate (a null over three orders falsifies P2-as-SOC)? and **(2) the magnon-BEC drive:** does a room-temperature YIG condensate's body-spanning spin phase respond to the x_v drive?

§13 The lattice-match substrate — first-principles material selection

This section is the **consonance lever of §3**: how closely the material's own geometry resonates with the lattice's. §§7–12 ranked materials by the *coupling* they offer; there is a deeper, first-principles way to ask it: the root papers say spacetime *is* a specific lattice — the `{5,3,3}` dodecahedral honeycomb. So which material's own structure most closely *reproduces* it? The closer the match, the higher the consonance fidelity f_{cons} (its ceiling $e^{-P/12} \approx 0.873$, `WARP_DRIVE_MATH_AUDIT.md` §10; the distinct quantity $e^{-P/24} \approx 0.934$ is the candidate translation coefficient of audit §9, not this ceiling). The full geometric argument — why pentagons cannot tile flat space (the frustration), why **dodecahedrane (C_{20})** is the literal primitive cell, Euler's forced-twelve-pentagons rule, and the dodecahedron ⇄ icosahedron duality — is derived in `the_periodic_table.html` and `shape_of_the_universe.html`; it is not repeated here. Only the conclusion it forces on *materials* is.

> **Derived · What the geometry picks out.** The exact lattice is forbidden in flat space (the crystallographic restriction theorem: 5-fold / icosahedral symmetry is non-periodic), so no element reproduces it — and that residual frustration is the `α√π` "almost-fit," the aliveness, not a defect. What the geometry *does* select splits along the duality: **carbon** in its curved degree-3 cages (fullerene / dodecahedral cage / nanotube) matches the *cell and the degree-3 vertex rule* exactly, but tiles only as one cage; **boron** (B_{12} icosahedra) and an **icosahedral quasicrystal** supply the long-range icosahedral *symmetry* and φ-scaling — the dual — at the cost of several edge lengths. Three independent roads (the cosmology, the periodic-table paper's "carbon = master joiner," and the §7 x_V ladder) converge on the same element; that convergence is the signal. Geometry-match and coupling-velocity are *different materials* — which is exactly the §8 layered disk. The makeable composition that follows is the one piece worth carrying in full.

Reasoning to the quasicrystal composition — the Tsai-type substrate

The carbon cage gives the exact *local* rule (degree-3, φ) but cannot tile; the other escape from the frustration above is to keep the exact *long-range* icosahedral symmetry and φ-scaling and relax the degree-3 / equal-edge rule — an **icosahedral quasicrystal (iQC)**. The striking result is that we can reason all the way to a specific, makeable composition.

The minimum is binary, not ternary — and not one

The famous iQCs are ternary (Al-Pd-Mn), but the floor is lower: stable **binary** icosahedral quasicrystals exist (i-$Cd_{5.7}Yb$, i-$Cd_{5.7}Ca$; and a whole binary *magnetic* family i-$RCd_{5.7}$, R = Gd…Tm). A genuine single-element *bulk* iQC does **not** exist, for a deep reason that is itself on-framework: quasiperiodicity needs *two* length scales in the irrational ratio φ, cleanest to imprint with two atom sizes; and metallic iQCs are stabilised by a Hume-Rothery pseudogap that sits at a precise electron-per-atom ratio e/a, which needs alloying to tune. So the "one atom coupling to three" ideal is unreachable in a bulk metal — **binary is the honest minimum** (the only "one-atom" quasicrystals are templated monolayers, e.g. Sb on an i-AgInYb surface).

Derived · The Tsai cluster is the atomic "dodecahedral cell." Tsai-type iQCs are built from concentric shells: an off-centre **tetrahedron** (core) → **dodecahedron (20 atoms, 12 pentagon faces)** → **icosahedron (12 atoms)** → icosidodecahedron (30) → rhombic triacontahedron (92). The dodecahedron shell *is* the pentagon-loop primitive cell (dodecahedrane) above; the 12-atom icosahedron shell is where the coupling-active atoms (rare-earth moments) sit — the **"12 ports" on the cell**; and the off-centre tetrahedral core is a built-in symmetry-breaking (phason) degree of freedom. The geometry the framework draws by hand is the geometry these atoms self-assemble into.

Layering in the coupling predicates → a specific composition

With the icosahedral+φ geometry (P1) fixed by the Tsai cluster, reason in the other predicates: P2 wants a heavy, strong-SOC atom; P3 wants a coherent order. Both are realised — and the order type is **continuously dialable by e/a** (spin-glass for e/a > ~1.9 → ferromagnetic ~1.57–1.9 → antiferromagnetic < ~1.57). The candidates:

Composition	P1 (ico+ϕ)	P2 (SOC/heavy)	P3 (coherent order)	N	note
i-$Au_{65}Ga_{20}Gd_{15}$	✓ Tsai	Au 5d + Gd 7 μ_B (L=0)	ferromagnet, T_C = 23 K (warmest iQC)	3	best overall
i-$Au_{65}Ga_{20}Tb_{15}$	✓ Tsai	Tb 4f, strong SOC; topological hedgehog texture	ferromagnet, T_C = 16 K + topological charge	3	best for SOC/topology (P2)
i-Tb-Cd / i-Dy-Cd ($RCd_{5.7}$)	✓ Tsai	Tb/Dy 4f SOC; Cd light	spin-glass ~5 K (frozen, not coherent)	2	fewest elements
i-$Au_{51}Al_{34}Yb_{15}$	✓ Tsai	Yb heavy, valence-fluctuation	quantum-critical (x, γ diverge T→0)	3	self-organised-critical electronic state
i-Al-Zn-Mg	✓	light, weak SOC	superconductor (P3), T_c ≈ 0.05–0.8 K	3	icosahedral + a real condensate (P1+P3) in one material; Mg-Zn(-Bi) appears in anomalous-materials

Composition	P1 (ico+ϕ)	P2 (SOC/heavy)	P3 (coherent order)	N	note
					reports — not discounted

> **Open — bench measures** · **The pseudogap is the catch — and it splits the optimum in two.** The same Hume-Rothery pseudogap that *stabilises* the icosahedral φ-structure (good for P1) *suppresses* itinerant conduction — iQCs are poor-conductor semimetals. So an iQC does *not* give a high-velocity electronic mode (low η, weak P4); its coherent channel must be the **localised-moment magnetic order or the SC condensate**, and it is cold (≤ 23 K). The geometry-match optimum (the iQC) and the coupling-velocity optimum (the high-η optical carbon/diamond aperture of §7) are therefore *different materials*. This is not a contradiction — it is the §8 layered disk made concrete: a **quasicrystalline icosahedral skeleton** (i-Au-Ga-Gd/Tb — the lattice-matched P1+P2+P3 register, with the 12 active ports on the cell) carrying a **high-η carbon/optical aperture** (the coupling surface). Reason to the geometry with the quasicrystal; reason to the coupling velocity with carbon; the disk wants both.

So your question answers cleanly: *yes*, you can reason to the exact characteristics and map them to elements — the geometry forces an icosahedral Tsai-type quasicrystal, "fewest elements" forces binary as the floor, and layering the coupling predicates picks out **i-$Au_{65}Ga_{20}Gd_{15}$** (or the Tb variant for spin–orbit and a topological spin texture, or binary i-Tb-Cd for minimal elements). The honest boundary remains: the iQC is the *geometry/symmetry* match (P1) with a real magnetic coupling channel (P2/P3), not the high-velocity electronic coupler — pair it with the carbon aperture for that. [E] the quasicrystal physics (binary iQCs, Tsai shells, the 2021 ferromagnetic and 2025 antiferromagnetic iQCs, e/a tuning, the pseudogap) is established and cited; [SPEC] the identification of the icosahedral shell with the lattice's "12 ports" is framework inference. Reproducible map: `hover/disk_material_ladder.py`.

§14 Driving the substrate — the conductivity question and the capacitor architecture

Everything above ranked *what the substrate is*; this section asks *how you energise it*, and the answer changes the material choice. If the drive is a high voltage applied across the substrate, the conductivity is suddenly decisive: a **conductor shorts** — the field collapses, current flows, and the stored drive energy dissipates as heat — whereas an **insulating dielectric between**

conductive electrodes holds the field. That is a capacitor, and the x_v drive energy becomes the stored electrostatic field energy

$$U_{\text{drive}} \;=\; \tfrac{1}{2}\,\varepsilon_r\,\varepsilon_0\,E^2\,V, \qquad u \;=\; \tfrac{1}{2}\,\varepsilon_r\,\varepsilon_0\,E_{\text{bd}}^2$$

with energy density u capped by the breakdown field E_{bd}.

> **Implementation consequence · The drive mechanism selects the substrate.** A *field/HV* drive wants an **insulator** (it holds the field); a *current/RF* drive of a magnetic order wants a **metal**. So the metallic magnetic quasicrystal of §13 (i-Au-Ga-Gd) is the right substrate *only if* you drive its magnetism with current/RF — under static HV it would short. If the disk is field-driven, the substrate must instead be a true insulating dielectric. The two are alternative drive architectures, not the same device.

14.1 Is there an insulating quasicrystal? (Mostly no — use a photonic one)

Bulk quasicrystal *alloys* are "bad metals," not dielectrics: even the most resistive (i-Al-Pd-Re, across a real metal–insulator transition) reaches only $\rho \approx 1\ \Omega\cdot\text{cm}$ — still ~16 orders of magnitude too leaky to hold off HV (diamond is $\sim 10^{16}\ \Omega\cdot\text{cm}$). The celebrated oxide quasicrystal (Ba-Ti-O, 12-fold) is a *single-atom interface monolayer* on Pt, not a bulk ferroelectric — a bulk ferroelectric quasicrystal does not exist. The realizable route to "insulating + 12-fold" is therefore a **photonic quasicrystal**: pattern a genuine dielectric (diamond, SiN, Si) in a 12-fold tiling, so the quasiperiodicity is *geometric* while the material stays a true insulator (complete photonic gaps are demonstrated even at low index). Your "insulating quasicrystal + graphene" becomes a **photonic-QC-patterned dielectric between graphene electrodes**.

> **Implementation consequence · A photonic quasicrystal may unify the two optima that §13 had to split.** A 12-fold/icosahedral photonic-QC-patterned *diamond* is simultaneously an *insulator* (holds the HV field), an *icosahedral/φ geometry register* (P1, by the pattern), and an *optical structure* carrying high-velocity EM modes ($\eta \approx c/n$ in the visible/IR — the high-η regime of §4, and the slow-light/band-edge modes of a photonic QC concentrate the field where the coupling is wanted) — all in the resilience/clean-field champion of §7. It does not supply the magnetic/spin channel (P2/P3); those still want a $BiFeO_3$ or magnetic layer. But "geometry-match vs η-match are different materials" (§13) softens here: the photonic-QC diamond is one structure that carries the icosahedral geometry *and* an optical-velocity mode at once.

14.2 The dielectric — energy density vs clean field-hold

The energy-density figure of merit *u* pulls against insulation quality: higher ε_r correlates with lower E_{bd} ($E_{bd} \propto E_g^{\sim 1.5}$), so ferroelectrics store more *nominal* energy but leak, age, and lose ε_r under bias, while wide-gap insulators store less but hold a *clean, loss-free* static field.

Dielectric	ε_r	E_{bd} (MV/cm)	u (J/cm³)	character
Diamond	5.7	~10 (→20)	~25 (→100)	cleanest high-field insulator; $\rho \sim 10^{16}$, $\kappa=22$ W/cmK
hBN	3.4	4–10	~4–10	atomically clean 2D insulator; graphene's partner
$BiFeO_3$ (multiferroic)	80–120	~2	~18	magnetoelectric coupling channel (P2); leaky → needs hBN buffer
$SrTiO_3$ (film)	~290	~6.8	~590*	best-behaved high-ε; *nominal
PZT relaxor / PZT-MgO	~1300	3.6–6.2	~68–169 (meas.)	highest u, but hysteretic/lossy — not a clean reservoir

14.3 The recommended stacks

Graphene is an excellent electrode here — transparent (97.7%/layer), patternable into the 12-segment rim drive, tunable work function (4.5–5.5 eV), and atomically smooth (no asperities → raises effective breakdown). Caveat: one atom thick (low current/ESD headroom), so use few-layer or graphene+metal-mesh on the HV bus bars. The two stacks that fall out are exactly the two threads of this paper:

- **Stack A — coupling channel:** `graphene / hBN / BiFeO₃ / hBN / graphene`. The $BiFeO_3$ core *is* the magnetoelectric (P2) coupling channel; the thin hBN buffers cap its leakage and present clean interfaces. Graphene-on-$BiFeO_3$ is already a demonstrated transparent- electrode multiferroic stack. Best when the order-parameter coupling is the point.
- **Stack B — clean field + geometry:** `graphene / hBN / diamond / hBN / graphene`, the diamond face etched as a 12-fold photonic quasicrystal. Diamond gives the cleanest, highest, loss-free static field *and* dumps the dielectric heating (κ=22 W/cm·K); the photonic-QC pattern injects the icosahedral/φ geometry into a true insulator — the only physically real route to "insulating + quasiperiodic." Best when a clean field reservoir and the geometry register dominate.

Architecture C — the spun chiral-vortex shell (the dynamic third option). Stacks A and B are *static* capacitors, and both lack the high-frequency / dynamic drive whose absence §15's vacuum nulls turn on ("no predicate P4"). The missing third architecture — reasoned in full in `viktor_schauberger.html` §9 from the Repulsine and the Tesla bladeless turbine, and routed here as a build note — makes the drive *dynamic by motion*: mechanically *spin* a charged stack whose electrodes are a *chiral spiral*, on a *closed domed boundary*, kept *cold*. The bulk rotation supplies the time-asymmetry and the rotating field a static plate cannot — *without needing a THz source* — which is exactly the spin-and-jerk **swirl** of `warp_drive.html` §4: a P4-bearing device by **motion** rather than frequency. Three additive moves over Stack A/B:

1. **Spin the capacitor** (the Repulsine; warp §4 spin) — bulk mechanical rotation of the charged A/B stack, adding the dynamic axis the static build cannot reach.
2. **Chiral-vortex electrodes, not concentric** (the Repulsine waveline; the ring-cascade of `all_you_need_is_water.html`) — a spiral electrode pattern gives the asymmetry κ *geometrically*, and the handed spiral matches the φ / icosahedral register the §13 lattice-match and the Stack-B photonic quasicrystal already want (the consonance lever).
3. **Boundary-layer coupling as a second channel** (the Tesla bladeless turbine, US 1,061,206) — a smooth / textured disc face that mechanically couples the spinning shell to the surrounding medium, a *mechanical* η lever alongside the optical-η one.

And **cold is a stated requirement, not an aesthetic:** the §8 phase-coherent condensate boundary is inherently cold, and Schauberger's cooling vortex is the everyday-water name for "keep the coherent mode un-decohered" (the §5b coherence lever). So Architecture C exercises *four* warp-drive levers in one device — spin (the swirl κ), stored charge (U_{drive}), the cold vortex (coherence), and the closed shell — that Stacks A and B, being static, do not.

> **Implementation consequence · Status: a build hypothesis for the bench, not a result.** Nothing here shows the Repulsine flew or that spinning yields net lift — that stays the same x_v bench question every architecture faces, and the §14.3 lemma below applies to C unchanged (the spinning shell *shapes* the drive; net translation still needs the x_v/lattice sink of §2, not the spin per se). What Architecture C changes is the *configuration to test*: not a static plate but a cold, spun, chiral-vortex charged shell with boundary-layer coupling — the one build that exercises the §4 spin/jerk/swirl levers a static capacitor cannot, assembled by Schauberger and Tesla 80–110 years early because they were copying nature and resonance. Right shape (mine it for the build); wrong physics if over-claimed (keep the conservation wall; demand the receipt).

> **Implementation consequence · Pais's patented geometry — mine the shapes, not the result.** The most detailed modern patents for this class of drive are Salvatore Pais's US Navy filings (**US**

10,144,532, the "inertial mass reduction device"; **US 10,322,827**, the "high-frequency gravitational wave generator"). Read them with the warp paper's register (`warp_drive.html` §6): the patents and their *geometry* are DOCUMENTED, but the device's efficacy is UNPROVEN — the Navy's own HEEMFG program (2016–19, ~$500k) found the "Pais Effect" "could not be proven," and its rig could not spin and vibrate at once, so the null is uninterpretable rather than a refutation. Strip the vacuum-polarization causal story and the patented *shapes* are real engineering motifs — three sharpen Architecture C and the §8 stack:

- **The double-shell resonant cavity** — a *charged* outer wall over an *insulated* inner wall. That is precisely the asymmetric two-shell capacitor on a closed boundary that §14 (the κ asymmetry), the Sport Model's insulation ring, and the §8 closed coherent shell already call for — now specified in a granted patent.
- **Counter-spinning cavities** — two resonant cavities that *counter-rotate*, a second opposed spin axis for stability and field-crossing that Architecture C (single rotation) did not carry.
- **Acousto-optic dual drive** — sound generators *and* microwave emitters vibrating the charged surfaces together: the *two-field* "sound writes the field the light reads" drive the sim lane independently built, and a concrete way to deliver the §4 spin/*jerk* by vibration, not bulk rotation alone.

And one convergence worth stating plainly: Pais's patent frames its effect as modifying "*the local spacetime … lattice energy density*" — the framework's exact object, reached in a granted US patent from the opposite (vacuum-engineering) direction. Take the geometry and the language as a fourth independent vantage on the one shape — alongside Schauberger, Tesla, and the Repulsine — but *not* the unproven "Pais Effect" as a result.

Open — bench measures · **The capacitor is a drive/coupling structure, not a propulsion mechanism — keep these separate.** Asymmetric dielectric capacitors (Biefeld–Brown "lifters") produce real thrust *in air* by ion wind, but every controlled *vacuum* test (Talley 1988; Campbell 2003; Tajmar 2004) found **null** net thrust; Buhler's mN-in-vacuum "Exodus Effect" is unreviewed and its "thrust persists after the voltage is removed" claim is a reactionless-drive red flag. The genuine field-asymmetry force is the Maxwell-stress / dielectrophoretic gradient $F = \nabla(\tfrac{1}{2}\varepsilon_r\varepsilon_0 E^2 \cdot V)$ — but on a closed, isolated device that is an *internal* stress that integrates to zero net translation (Newton's third law, again). So the HV capacitor is the right way to *store and shape the drive field and couple it to the substrate's order*; it does **not** by itself give net lift. Net translation still requires the x_v/lattice momentum sink of §2 — the capacitor energises the coupling, the lattice carries the reaction.

14.4 Propellantless, not reactionless

The line this whole programme walks — a **propellantless** drive (no ejected reaction mass) is allowed; a **reactionless** one (a force with no reaction at all) violates momentum conservation and is not — is stated in full in `WARP_DRIVE_MATH_AUDIT.md` §2 (the two-regime split of §2 above: gradient force on a separate object is regime (a) and conservation-safe; net self-propulsion is regime (b), gated by the open lattice-mode speed v_L via $\langle F \rangle \lesssim \langle P_{coupled} \rangle / v_L$, and hover against Earth's field is the conservation-safe static-equilibrium case). The only materials consequence is the one the §14.3 lemma already draws: the HV capacitor *stores and shapes the drive field and couples it to the substrate's order*; it is not itself the propulsion. Net translation needs the lattice momentum sink — the capacitor energises the coupling, the lattice carries the reaction. The honest test is always the single question *where does the reaction momentum go?*; for this disk the answer is "the lattice," and whether that coupling carries momentum (and at what v_L) is the open bench question the whole substrate hunt serves.

§15 Cross-check against documented electrostatic-thrust devices

A force law earns trust by surviving contact with published data. The closest documented devices to the §14 capacitor are the electrostatic thrusters — ion-wind "lifters" and the controlled vacuum/sealed nulls. They let us do two things: validate the *conventional* part of our math against measured thrust, and let the published *nulls* put a real upper bound on the open coefficient $C_{sub} \cdot |x_v|^2$. To be explicit about provenance: every null in this section is a **published literature measurement** (Tajmar 2004; Bartlett & Tajmar 2021; Talley 1991), used here as bounds — **the project's own bench has not yet measured any substrate**, so nothing in this table is a programme-internal result. Reproducible: `hover/capacitor_cross_check.py`.

device	V	medium	thrust (obs/bound)	cross-check result
Bahder/Fazi, Canning lifters	30–40 kV	air	~60 mN (measured)	ion-wind $T = I \cdot d/\mu$ reproduces it (×0.1–0.8 geom.) ✓
Tajmar 2004 (corona-excluded)	38 kV	sealed	< 10 μN [peer-rev.]	$C_{sub} \cdot |x_v|^2 < \sim 2 \times 10^{-3}$
Bartlett & Tajmar 2021 (dielectric cap)	≤ 10 kV	balance	~nano-N [peer-rev.]	$C_{sub} \cdot |x_v|^2 < \sim 2 \times 10^{-10}$

device	V	medium	thrust (obs/bound)	cross-check result
Buhler patent (US11511891B2 / WO)	25 kV	**air**	237 μN (US) / 237 mN (WO)*	his own $F=\varepsilon_0[E^2A]$ = internal Maxwell stress; measured in air; no vacuum test

> **Derived · Conventional math: validated. xv coefficient: bounded, not refuted.** The ion-wind relation reproduces the measured air thrusts, and the Maxwell-stress plate force is the legitimate electrostatic force — but it is *internal* (electrode↔electrode), so it gives zero net translation, which is exactly why the vacuum tests null. The tightest published null (Bartlett's dielectric capacitor) bounds $C_{sub} \cdot |x_v|^2 < \sim 2\times10^{-10}$. For scale: that bound still sits ~33 orders of magnitude *above* the bare-vacuum PV floor $8\pi G/c^4 \approx 2\times10^{-43}$ (`WARP_DRIVE_MATH_AUDIT.md` §8) — so the published nulls bound the coherent-enhancement claim (the single open bet) without touching the floor, which is ordinary general relativity.

Now compare that bound to the framework's naive estimates (§6, §13): an *acoustic* substrate sits at $\sim10^{-14}$ — far below the bound, so the null cannot see it (consistent). An *optical* substrate's naive $|x_v|^2 \sim 6\times10^{-2}$ sits ~8 orders *above* the bound — which looks like a problem until you notice **the null devices are statically-driven capacitors with no high-frequency mode — no predicate P4.** The framework's whole claim is that the high-η optical channel only switches on under THz-to-optical drive; a static HV capacitor structurally cannot access it. So the static-cap nulls rule out a large *static* x_v coupling (which the framework never claims) and say nothing about a high-frequency-driven coherent substrate (which it does require). The apparent tension is resolved by the framework's own P4 requirement — the cross-check is self-consistent.

> **Implementation consequence · The Buhler empirical case lives in the mechanism paper.** Buhler's patent gives F ∝ (geometry) (dielectric constant) (V^2), which matches our $U_{drive} = \frac{1}{2}\varepsilon_r\varepsilon_0 E^2 V$ scaling; his reported "persists-after-disconnect" is the stored-energy signature an F ∝ U_{drive} drive predicts; and his patent explicitly names *no reaction sink* — precisely the piece this framework supplies (the lattice). The full treatment — the ~2,000 claimed vacuum runs, the double-edged "persists-unplugged" claim, the chamber-variation control that would discriminate a lattice coupling from a trapped-charge artifact, and the DC-vs-frequency lever Buhler has not explored — is in `warp_drive.html` §6–§7 and is not duplicated here. What this paper keeps is the part that bears on *materials*: the null-bound above (an upper limit on the coupling) and the quantitative number to beat (below).

15.3 The quantitative check — Buhler's number against the force law

Buhler's data is the only place to ask the sharpest question: *does the force law reproduce a real (claimed) number with a sane coupling, or does it break?* The usable figure is F ≈ **5–10 mN** (his vacuum-*claimed* stacked result — the patent's 237 μN in §15's table was measured *in air* and so cannot be separated from ion wind), at **40 kV DC**. The quantities $\frac{1}{2}CV^2$ needs — capacitance, K, area, gap — are *not published*, so the capacitor must be estimated and this is an order-of-magnitude check, not a fit (`hover/buhler_cross_check.py`).

> **Derived · The force law does not break on Buhler's number, and the scaling agrees.** For a plausible 40 kV element (PTFE, ~7 cm², 0.5 mm gap → C ≈ 26 pF, U = $\frac{1}{2}CV^2$ ≈ 21 mJ, L_{grad} ≈ 3 cm), the law $F = (C_{sub}|x_v|^2 \cdot \kappa) \cdot U/L$ back-solves to an implied coupling $\mathbf{C_{sub}|x_v|^2 \cdot \kappa \approx 2\times10^{-4}}$. The one structural test passes: Buhler reports forces that *stack linearly* (N layers → N× force), which is exactly the force law's linearity in stored energy — the per-element coupling is the same read from a single element or the stacked total.

> **Open — bench measures · Where it lands — and three caveats that keep it honest.** $\sim 2\times10^{-4}$ is **~125× below this paper's hover threshold** (§6, $\sim 3\times10^{-2}$) and ~1000× below the EM-optimal estimate (~0.25) — *but Buhler runs DC*, so under hypothesis A (frequency is a lever) a small coupling is the predicted shape and the ~125× is what the §7 sweep would buy back, while under hypothesis B (DC suffices) $\sim 2\times10^{-4}$ is simply the genuine coupling and the gap to hover is real. The §7 fork decides which. The caveats: (i) one data point cannot *validate* a two-parameter law, only fail to break it; (ii) the coupling scales as 1/C and C is our estimate, so it is good to ~±10×; (iii) the force law is written for a *driven coherent mode*, and a static DC field is not obviously that — so Buhler's effect, if real, is either a static special case of the law or a different mechanism, which is the fork warp-drive §7 resolves. **Consistent within an order of magnitude; not validated.**

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44. Itoh, K.M. & Watanabe, H. (2014). "Isotope engineering of silicon and diamond for quantum computing and sensing applications." *MRS Bull.* 39, 208; Balasubramanian, G. *et al.* (2009). "Ultralong spin coherence time in isotopically engineered diamond." *Nat. Mater.* 8, 383. (^{28}Si / ^{12}C coherence — the coherence lever, §5b.)

45. Schindler, F. *et al.* (2018). "Higher-order topology in bismuth." *Nat. Phys.* 14, 918. (Bi as a topological semimetal and strong diamagnet; condensate-boundary layer, §8.)

46. Vallée, J., Nolan, G.P. *et al.* (2022). "Improved instrumental techniques, including isotopic analysis, applicable to the characterization of unusual materials." *Prog. Aerospace Sci.* 128, 100788; AARO *Historical Record Report Vol. 1* (DoD, 2024) with the ORNL metallic-specimen analysis. (Materials reverse-engineering efforts — results terrestrial; §5b, §8.)

47. Pais, S.C. (US Navy / NAWCAD). US Patent 10,144,532 B2, "Craft using an inertial mass reduction device" (2018); US 10,322,827 B2, "High frequency gravitational wave generator" (2019). (The patented *geometry* mined in §14.3: double-shell charged/insulated resonant cavity, counter-spinning cavities, acousto-optic dual drive; the patent's own "local spacetime

… lattice energy density" language. *Documented patents; device efficacy unproven — HEEMFG null, 2016–19; mine the shapes, not the result*.)

Draft engineering/materials paper in the vertex-sharing research program (earlier drafts wrote the coupling x_v; current objects: $g \leq 1/\sqrt{3}$, $x_{sub} = \Sigma g_v \psi^*\psi$, S_{coh}, f_{cons}, κ). Its role is **substrate selection**; mechanism questions belong to `the_shape_of_gravity.html` and `WARP_DRIVE_MATH_AUDIT.md`. It ranks substrates against four material-set levers (energy ceiling, coherence, consonance, coupling; the Steiner ether names of earlier drafts are provenance only, §3), of which three are sound and the coupling's frequency-scaling is the open A/B fork; it does not claim a measured force. Quantities are tagged derived / estimate / open / bench; the open numbers (η multipliers, C_{sub}, where a body sits on the $S_{coh} \cdot f_{cons}$ coherence dial) are bounded so far only by published literature nulls — the project's own bench has not yet measured any substrate. Reproducible numbers: `hover/disk_material_ladder.py`.